



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	PHYSICAL CHEMISTRY –II: CHEMICAL THERMODYNAMICS, KINETICS AND SURFACE CHEMISTRY				
Type of Course	MAJOR				
Semester	IV				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	[Prerequisites: NCERT or equivalent chemistry syllabus of XI and XII, https://onlinecourses.swayam2.ac.in/nce24_sc07/preview Fundamentals of Chemical Thermodynamics. Path function and state function - Thermodynamic terms for defining System - Surroundings - Types of systems]				
Course Summary	We witness, feel and create physical and/or chemical change(s) everyday. What drives these changes? This course deals with the principles of chemical thermodynamics and chemical kinetics to answer these questions. The subject matter covered will enable the student to understand the relation between heat, work, temperature, and energy. Here, the various tools to evaluate chemical systems in equilibrium and rates of chemical reactions are also introduced. Further, the concept of catalysis and its importance in industrial processes is also included in this course.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To Understand the fundamental concepts of	U	F	Assignments/Quiz/Seminars

	thermodynamics and identify it with the real world			
CO2	To apply thermochemical principles to chemical reactions	Ap	C	Work out problems/assignments/Test
CO3	To apply the concept of kinetics and catalysis to various chemical and physical processes	Ap	C	Work out problems/assignments/Test
CO4	To interpret kinetic data using graphical representations and evaluate the rate of a reaction	An	P	Quiz/Discussion
CO5	To evaluate the surface area of catalysts using various adsorption isotherms	Ap	P	Quiz/Discussion
CO6	To apply the theories of kinetics and adsorption through laboratory experiments	C	P	Lab work/Viva voce exams
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs (45+30)	Marks
I	FIRST LAW OF THERMODYNAMICS AND THERMOCHEMISTRY		15	33
	1	Intensive and extensive properties - Steady state and equilibrium state. Concept of thermal equilibrium	1	
	2	Zeroth law of thermodynamics. Intensive, extensive and state variables (state functions), Introduction to partial derivatives and line integrals, Euler theorem, Exact and Inexact	3	

		differential, Illustration of exact differential using molar volume of ideal gas.		
	3	First law of thermodynamics – Concept of heat (q), work, internal energy(U) and enthalpy (H) - Heat capacities at constant volume and at constant pressure & their relationship - Expansion of an ideal gas under isothermal and adiabatic conditions - Work done in reversible isothermal and adiabatic expansion.	4	
	4	Joule-Thomson effect- significance of term $(dU/dV)_T$ - Liquefaction of gasses - Derivation of the expression for Joule Thomson coefficient – Inversion temperature.	3	
	5	Thermochemistry: Heat changes during physical and chemical changes. Hess's Law.	2	
	6	Temperature dependence of reaction enthalpies- Kirchoff's law. Bond dissociation energies. Resonance energy from thermochemical data.	2	
II	SECOND & THIRD LAWS OF THERMODYNAMICS		10	21
	7	Limitations of first law and Need for the second law – Kelvin and Clausius statements. Carnot's theorem and Heat engine and its efficiency.	2	
	8	Concept of Entropy. Calculation of entropy change for reversible and irreversible processes. Statement of first law in terms of entropy. Entropy change during the isothermal mixing of ideal gasses.	2	
	9	Energy functions (Gibbs free energy (G) and Helmholtz energy (A)) and their variation with T and P.	2	
	10	Maxwell's relations. Gibbs-Helmholtz equation - Criteria for spontaneity and equilibrium - Significance of Clausius inequality.	2	
	11	Partial molar free energy - Concept of chemical potential - Gibbs-Duhem equation.	1	
	12	Third law of thermodynamics - Nernst heat theorem - Statement of third law.	1	
III	CHEMICAL KINETICS		15	33
	13	Rate of a reaction - Factors influencing the rate of a reaction- Concentration, Temperature, Surface area and Catalyst - Rate law - Order and molecularity -	2	
	14	Derivation of rate constants for first, second (with same and different reactants), third (with same reactants only) and zero	4	

		order reactions with examples. Data interpretation including graphical representations		
	15	Half-life period (derivation for first and n^{th} order reactions) - Methods to determine the order of a reaction.	1	
	16	Effect of temperature on reaction rates - Arrhenius equation - Determination and significance of Arrhenius parameters	1	
	17	Theories of reaction rates - Collision theory - Derivation of rate equation for bimolecular reactions using collision theory - Transition state theory - Expression for rate constant based on equilibrium constant and thermodynamic aspects – Eyring equation (derivation not required)	5	
	18	Unimolecular reactions - Lindemann mechanism.	2	
IV	SURFACE CHEMISTRY, ADSORPTION AND CATALYSIS		5	11
	19	Solid surfaces, microstructure and elementary idea about microscopic techniques for studying the surface of solids (SEM, TEM, STM, AFM)	1	
	20	Physisorption, Chemisorption. Adsorption isotherms – Langmuir, Freundlich and BET (No derivation required). Determination of Surface area, Particle size and surface area, Activated charcoal and its uses	2	
	21	Homogeneous and heterogeneous catalysis - Theories of homogenous and heterogenous catalysis with examples	1	
	22	Enzyme catalysis - Michaelis-Menten equation (derivation not required). Application of enzyme technology for environmental, medical, agricultural, and industrial benefits.	1	
V	PHYSICAL CHEMISTRY- PRACTICALS-2		30	
		A minimum of 5 practical experiments out of which TWO EACH from sections 1 and 2 must be performed and reported. For plots/graphs, suitable softwares may be used and printed hard copies may be presented. Practical records may be in handwritten or computer-typed printed form.		
		Section 1		
		1. Determination of rate constant of the Acid Hydrolysis of ethyl acetate	3	
		2. Determination of effect of temperature on the rate of acid hydrolysis of ethyl acetate	3	
		3. Determination of order of the reaction between crystal violet dye and NaOH (or Fuchsin and NaOH) by using	3	

		a colorimeter/spectrophotometer 4. Kinetics studies of reaction between KMnO_4 and Oxalic acid 5. Open ended Section 2 6. Adsorption of oxalic acid on activated charcoal and thereby determining the adsorption isotherm. 7. Observation of decolourisation of a suitable dye on activated charcoal or filter paper via visual or colorimetry/spectrophotometry 8. Verification of Hess's law by using Mg, MgO and HCl reactions. 9. Effect of Mn^{2+} catalyst on reaction kinetics of KMnO_4 vs Oxalic acid 10. Open ended	3 3 3 3 3 3	
		References Module I and II 1. Physical Chemistry: Thermodynamics, Structure and Change, 10th Edition, P. Atkins and J. de Paula, (W. H Freeman and Company, New York) 2. D. A. McQuarrie, J. D. Simon, Physical Chemistry – A Molecular Approach, (Viva, 2001.) 3. T. Engel, P. Reid, Thermodynamics, Statistical Thermodynamics & Kinetics, Pearson Education, Inc: New Delhi, 2007. 4. J. Rajaram, J. C. Kuriacose, Chemical Thermodynamics, Pearson Education, New Delhi, 2013. Module III and IV 5. Physical Chemistry: Thermodynamics, Structure and Change, 10th Edition, P. Atkins and J. de Paula, (W. H Freeman and Company, New York) 6. K. Laidler, Chemical Kinetics, 3rd Edn., Pearson Education, New Delhi, 2004. Module V 7. Findlay's Practical Physical Chemistry, Ninth Edition, Revised and Edited by B P Levitt, (Longman, London, 1973)		

		8. Advanced Physical Chemistry: Practical Guide, C. Arora and S. Bhattacharya, Bentham Books, UAE, 2022 9. J. B. Yadav, Advanced Practical Physical Chemistry, Goel Publications, Meerut, 2008.. 10. R. C. Das, B. Behra, Experiments in Physical Chemistry, Tata McGraw Hill, New Delhi, 1983 Further reading 11. F. Daniels, R. A. Alberty, Physical Chemistry, 5th Edn., John Wiley and Sons, Canada, 1980. 12. I. B. R. Puri, L. R. Sharma, M. S. Pathania, Principles of Physical Chemistry, 46th Edn., Vishal Publishing Company, New Delhi, 2013. 13. G. M. Barrow, Physical Chemistry, 5th Edn., Tata McGraw Hill Education, New Delhi, 2006. 14. F. Daniels, R. A. Alberty, Physical Chemistry, 5th Edn., John Wiley and Sons, Canada, 1980. 15. D. P. Shoemaker, C. W. Garland, Experiments in Physical Chemistry, McGraw-Hill Book Company, New York, 1962. 16. W. G. Palmer, Experimental Physical Chemistry, Cambridge University Press, Cambridge, 2009		
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Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	-	-	3	2	3	2	2	-	2	-	1
CO2	3	2	-	-	3	2	3	2	1	-	2	-	1
CO3	3	2	-	-	3	2	3	2	1	-	2	-	1
CO4	3	2	-	-	3	3	3	2	1	-	1	-	1

CO 5	3	2	-	1	3	3	3	2	1	-	3	-	1
CO 6	3	-	2	3	3	3	3	2	1	2	3	2	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment /viva	Practical skill evaluation	End Semester Examinations
CO 1		✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4	✓			✓

CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓